Project Report: Student Burnout Prediction Using Time-Series + Behavioral Data

Group Members

Nishat Tasnim Luba (24241354) (section -01)
Nazifa Tahsin (22201713) (Section-03)
Nabilah Hossain (22299295) (section-04)
Tasneem Tuhfa (24241172) (section -04)

Results of the code:

RandomForestClassifier:

```
==============================
Model: Random Forest
Accuracy:  0.6562
Precision: 0.6645
Recall:    0.9752
F1 Score:  0.7904
ROC-AUC:   0.5046
```

Logistic Regression:

```
==============================
Model: Logistic Regression
Accuracy:  0.4990
Precision: 0.6657
Recall:    0.4953
F1 Score:  0.5680
ROC-AUC:   0.4997
```

KNN:

```
Model: KNN
Accuracy:  0.4901
Precision: 0.6631
Recall:    0.4739
F1 Score:  0.5527
ROC-AUC:   0.5011
```

Decision Tree:

```
==============================
Model: Decision Tree
Accuracy:  0.5484
Precision: 0.6666
Recall:    0.6419
F1 Score:  0.6540
ROC-AUC:   0.5025
```

MLP:

```
==============================
Model: MLP
Accuracy:  0.6013
Precision: 0.6641
Recall:    0.8102
F1 Score:  0.7299
ROC-AUC:   0.4993
```

Naive Bayes:

```
==============================
Model: Naive Bayes
Accuracy:  0.5022
Precision: 0.6601
Recall:    0.5183
F1 Score:  0.5806
ROC-AUC:   0.4962
```

Gradient Boosting:

```
==============================
Model: Gradient Boosting
Accuracy:  0.6377
Precision: 0.6644
Recall:    0.9194
F1 Score:  0.7714
ROC-AUC:   0.4990
```

AdaBoost:

```
==============================
Model: AdaBoost
Accuracy:  0.5260
Precision: 0.6640
Recall:    0.5812
F1 Score:  0.6198
ROC-AUC:   0.4982
```

SVM Linear:

```
==============================
Model: SVM Linear
Accuracy:  0.4876
Precision: 0.6682
Recall:    0.4556
F1 Score:  0.5418
ROC-AUC:   0.5022
```

SVM RBF:

```
==============================
Model: SVM RBF
Accuracy:  0.5370
Precision: 0.6684
Recall:    0.6027
F1 Score:  0.6339
ROC-AUC:   0.5071
```

XGBOOST:

```
==============================
Model: XGBoost
Accuracy:  0.6602
Precision: 0.6651
Recall:    0.9851
F1 Score:  0.7940
ROC-AUC:   0.5051
```

Best Model: XGBOOST

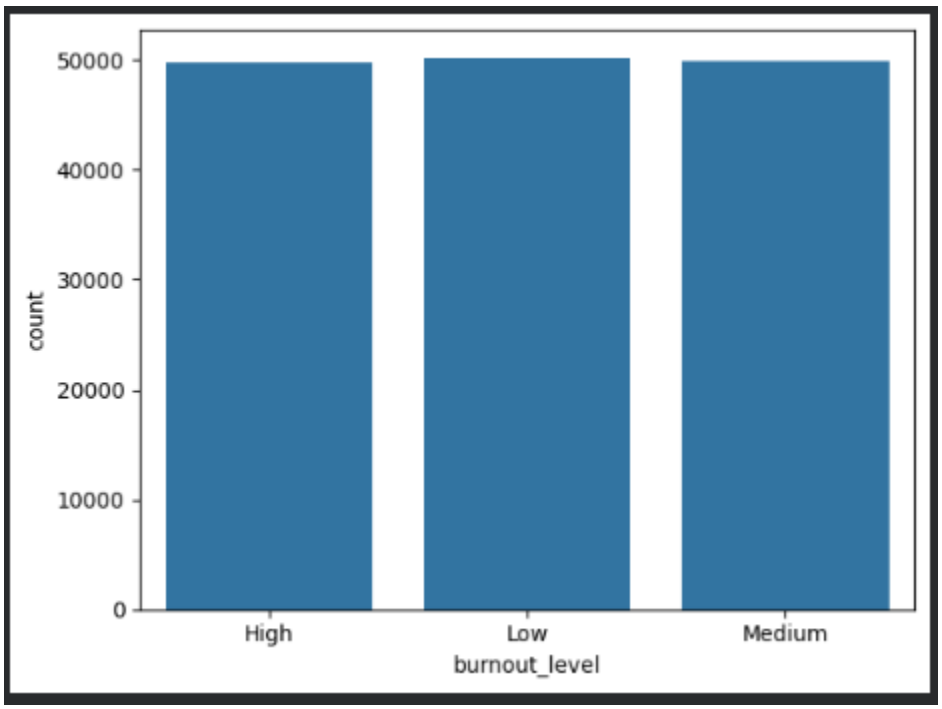
	Model	Accuracy
10	XGBoost	0.660233
3	Random Forest	0.656200
6	Gradient Boosting	0.637667
4	MLP	0.601300
2	Decision Tree	0.548433
9	SVM RBF	0.537033
7	AdaBoost	0.525967
5	Naive Bayes	0.502233
0	Logistic Regression	0.499033
1	KNN	0.490067
8	SVM Linear	0.487600

	Accuracy	Precision	Recall	F1 Score	ROC-AUC
XGBoost	0.660233	0.665076	0.985060	0.794043	0.505073
Random Forest	0.656200	0.664549	0.975184	0.790443	0.504635
Gradient Boosting	0.637667	0.664432	0.919386	0.771389	0.499036
MLP	0.601300	0.664078	0.810197	0.729896	0.499261
Decision Tree	0.548433	0.666615	0.641851	0.653998	0.502463
SVM RBF	0.537033	0.668409	0.602697	0.633854	0.507113
AdaBoost	0.525967	0.663975	0.581190	0.619831	0.498181
Naive Bayes	0.502233	0.660069	0.518273	0.580640	0.496183
Logistic Regression	0.499033	0.665701	0.495262	0.567971	0.499683
KNN	0.490067	0.663066	0.473856	0.552716	0.501125
SVM Linear	0.487600	0.668186	0.455607	0.541791	0.502168

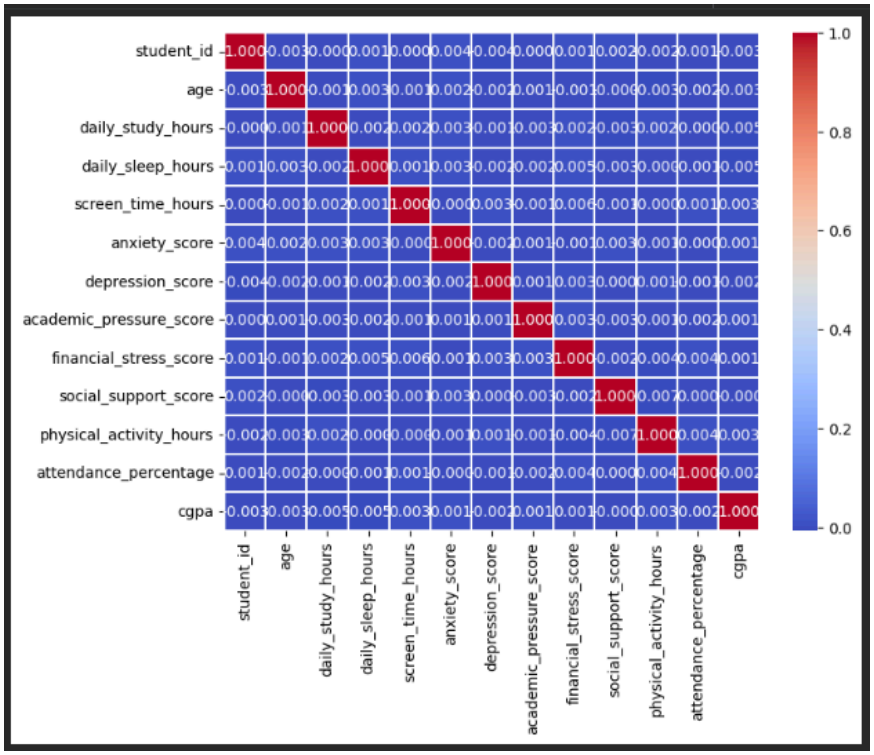
Figures:

EDA:

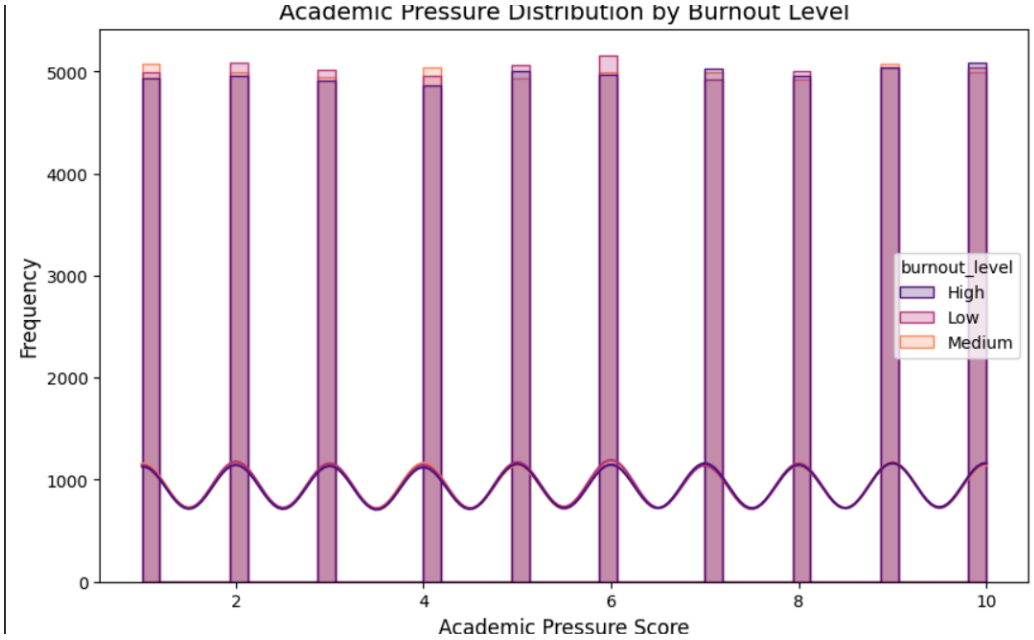
Countplot:



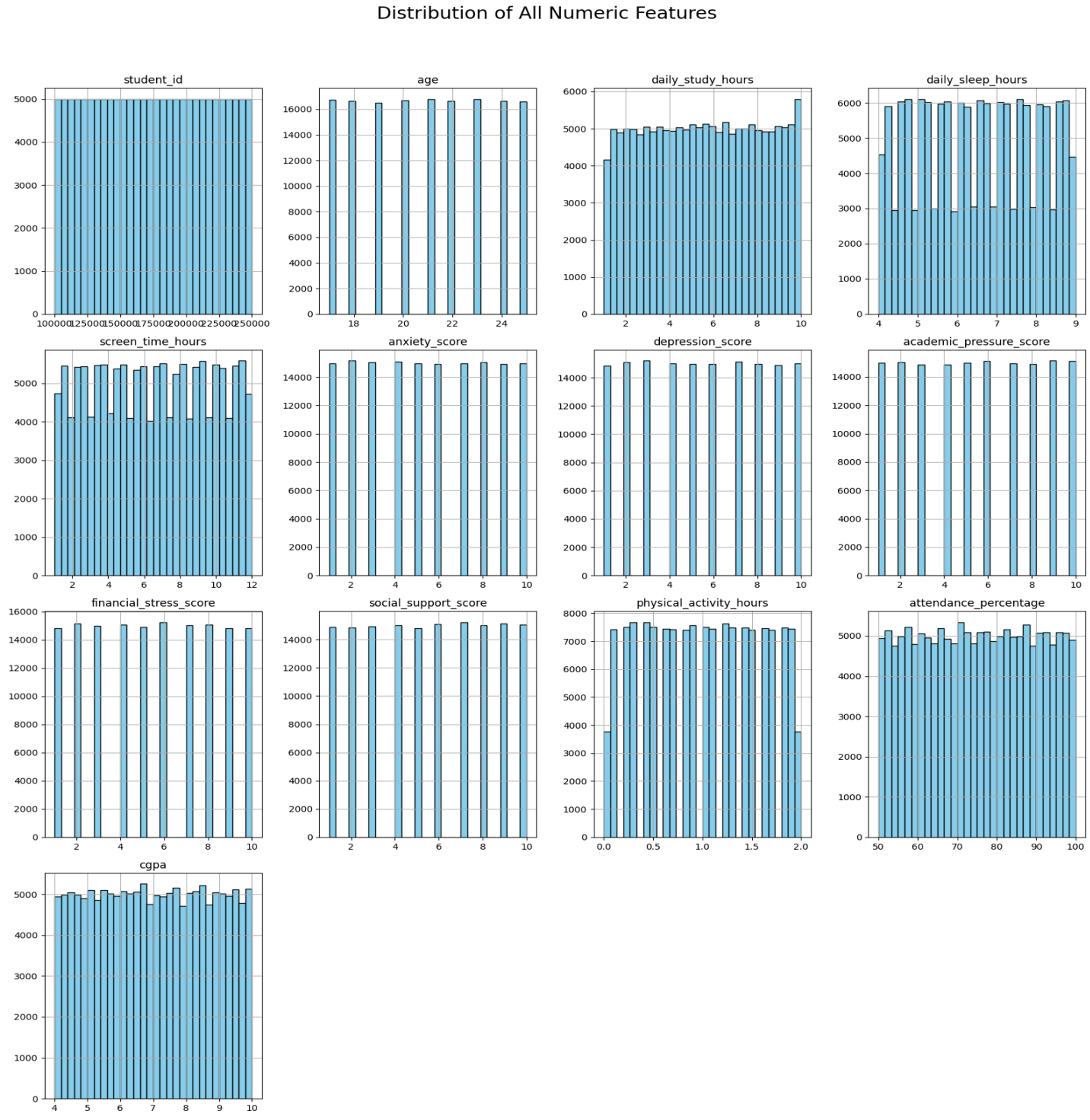
Correlation Heatmap:



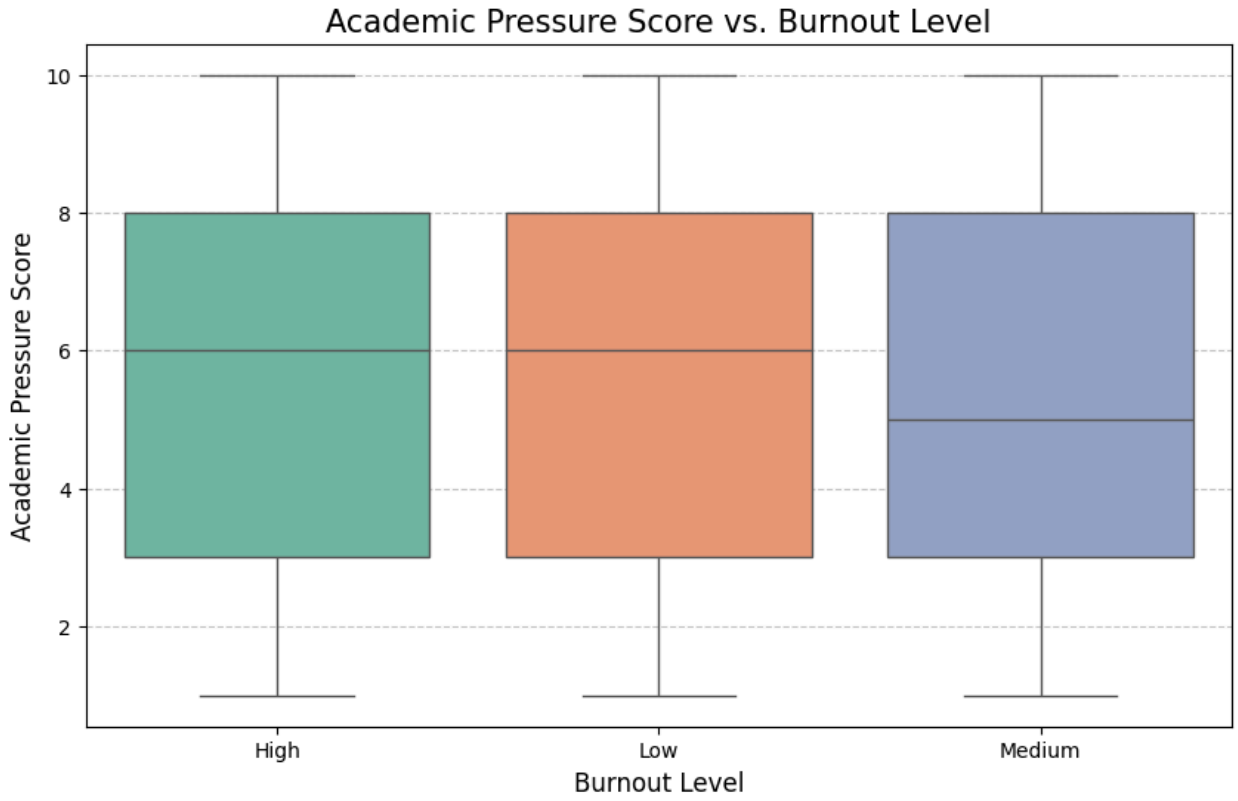
Hue Histogram:



Grid Histogram:

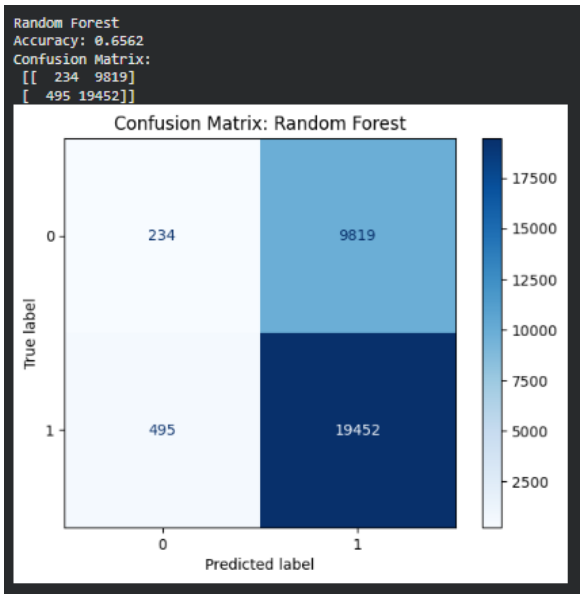


Boxplot:

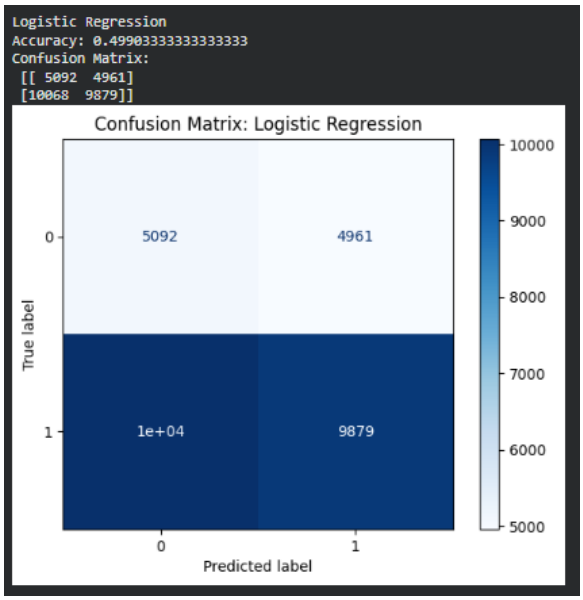

Academic Pressure Score vs. Burnout Level
Academic Pressure Score
10
8
6
4
2
High
Low
Medium
Burnout Level

Confusion matrix:

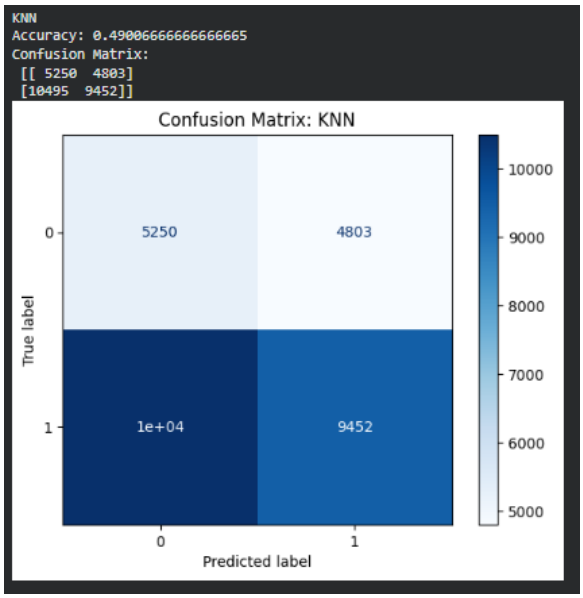
RandomForestClassifier:



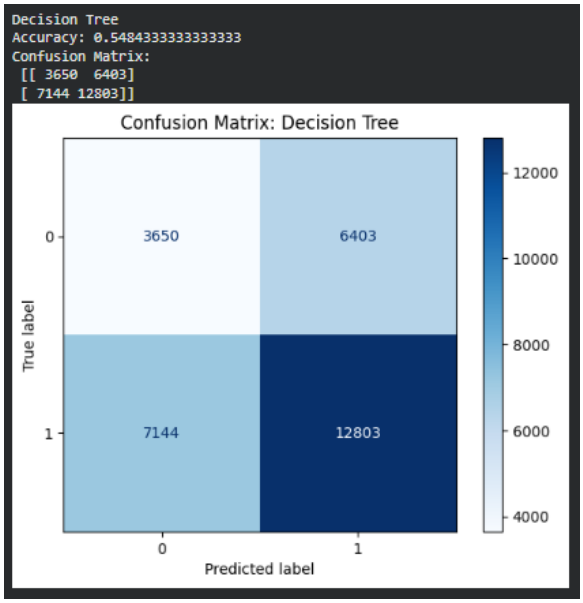
Logistic Regression:



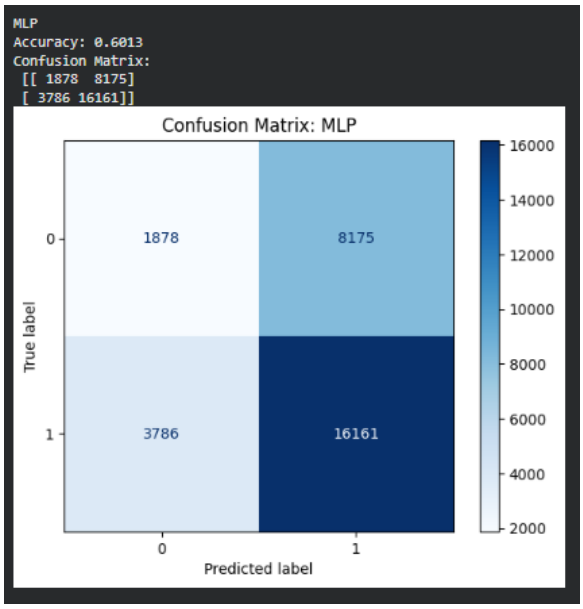
KNN:



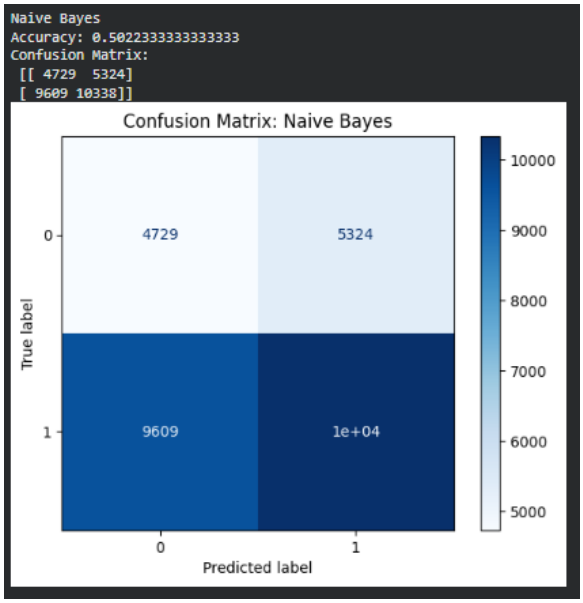
Decision Tree:



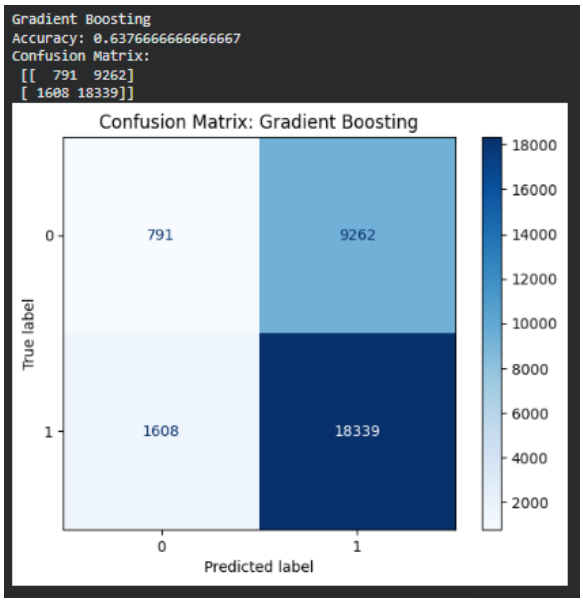
MLP:



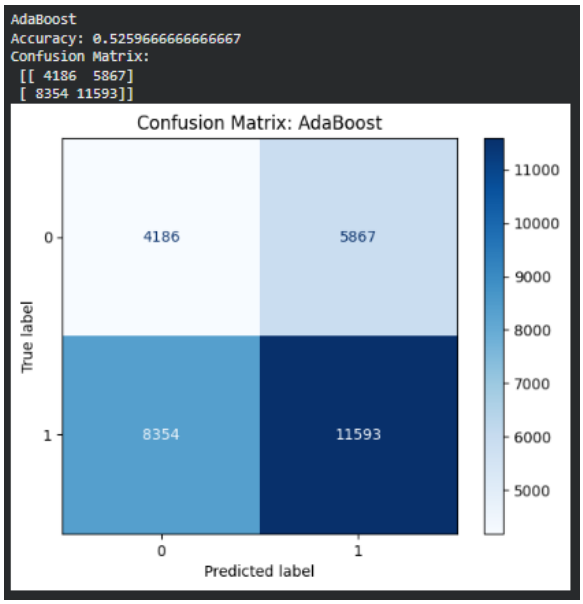
Naive Bayes:



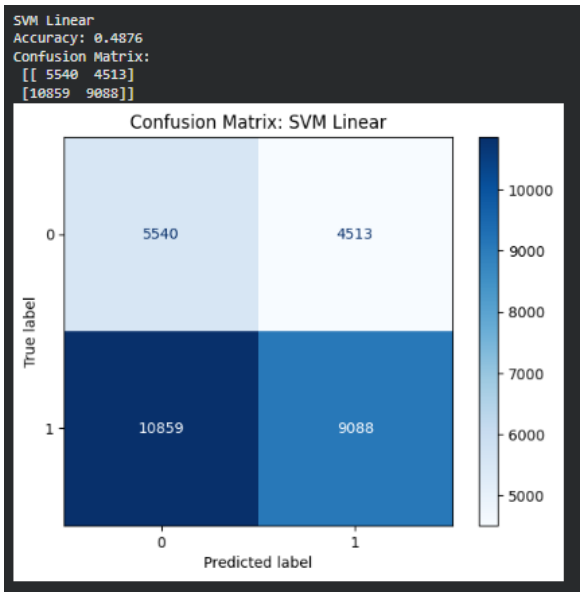
Gradient Boosting:



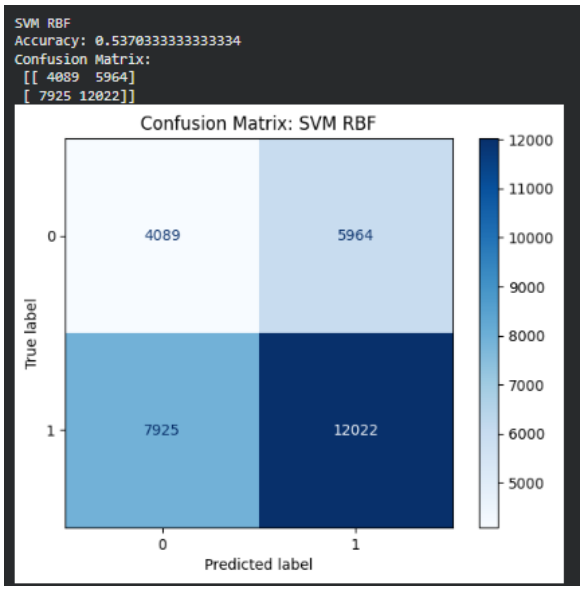
AdaBoost:



SVMLinear

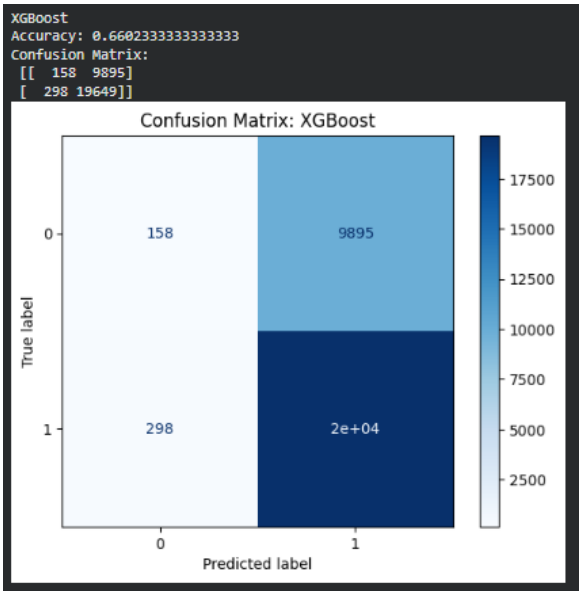


SVM RBF



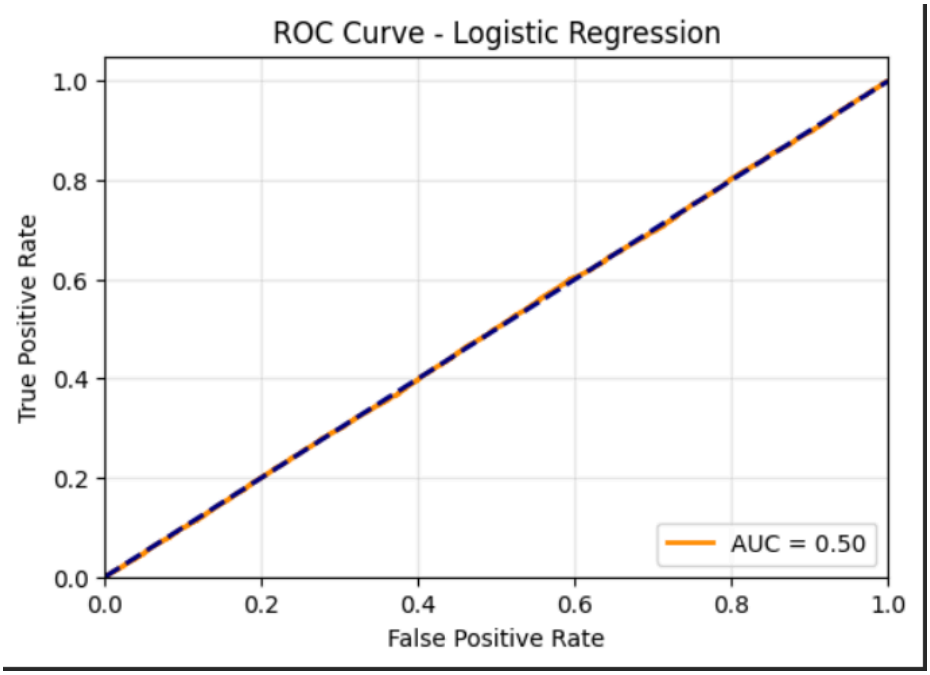
XG Boost

```
XGBoost
Accuracy: 0.6602333333333333
Confusion Matrix:
 [[  158  9895]
 [  298 19649]]
```

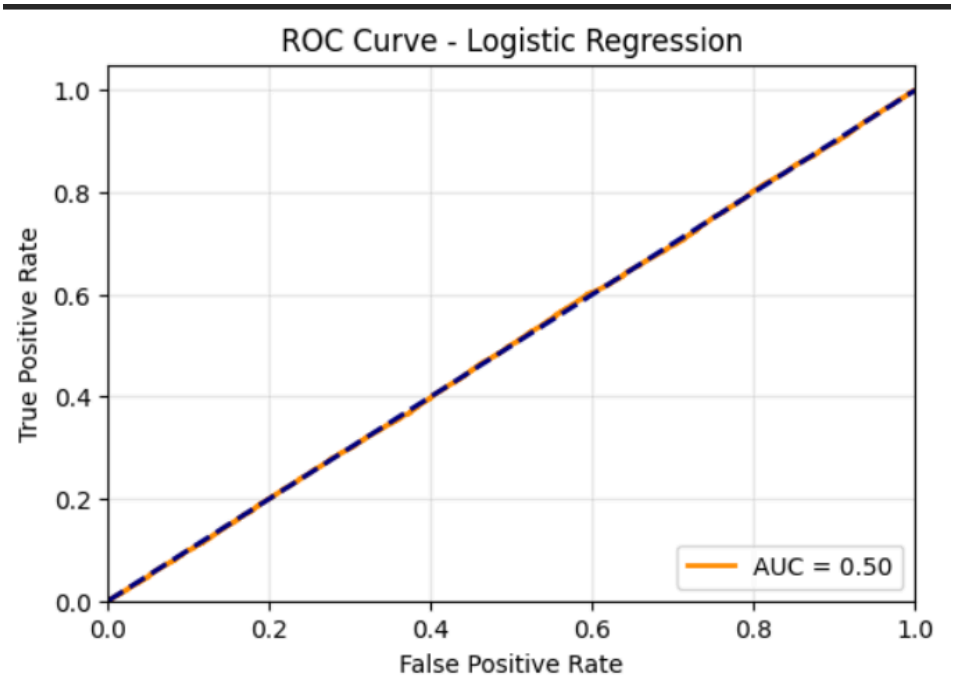


ROC CURVE:

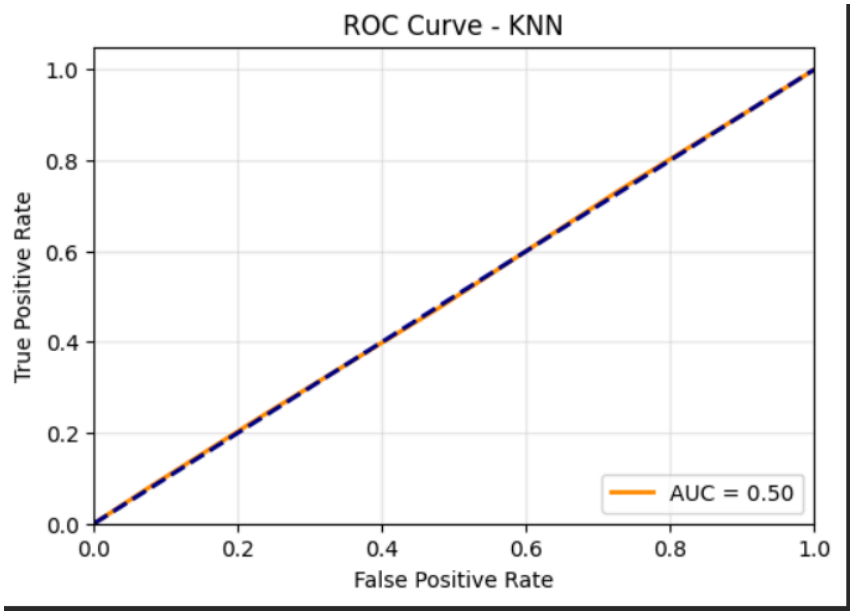
RandomForestClassifier:



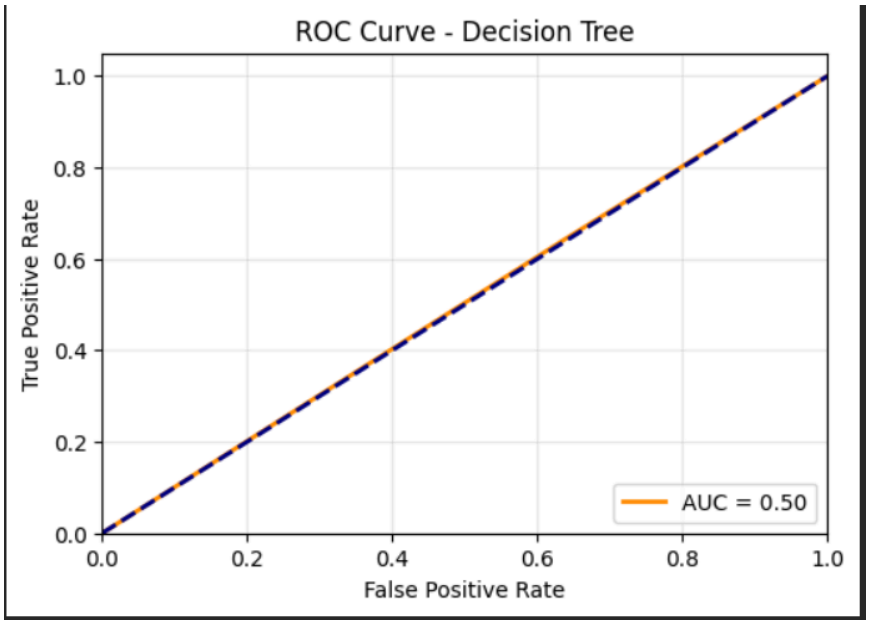
Logistic Regression:



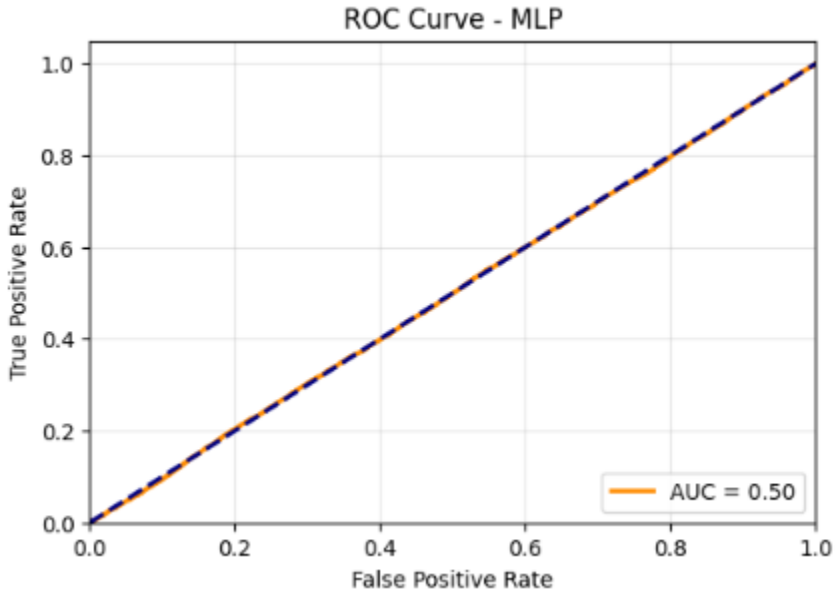
KNN:



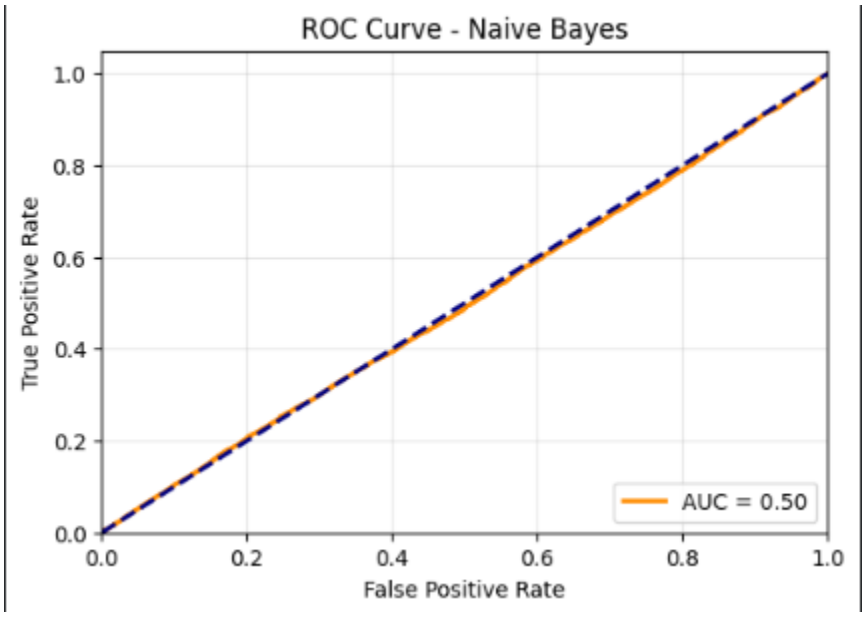
Decision Tree:



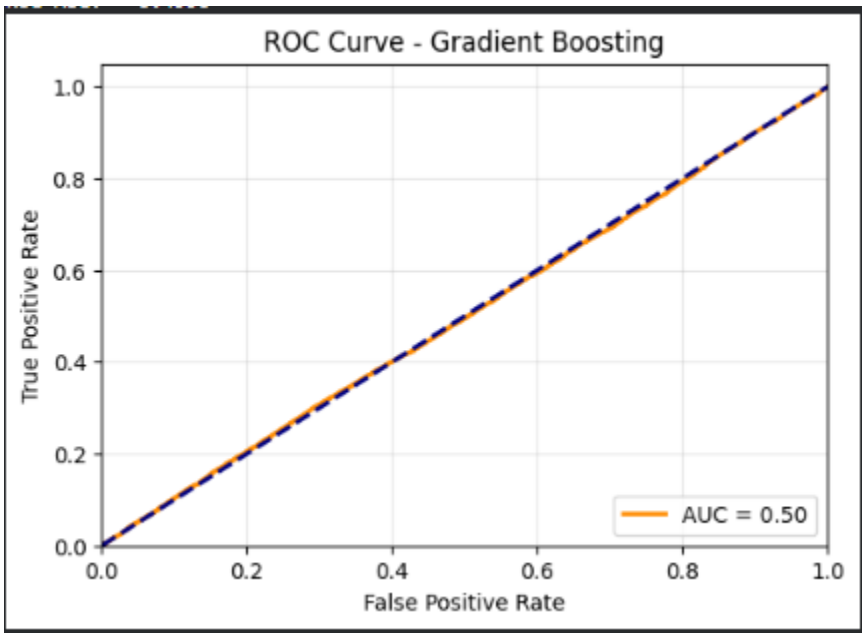
MLP



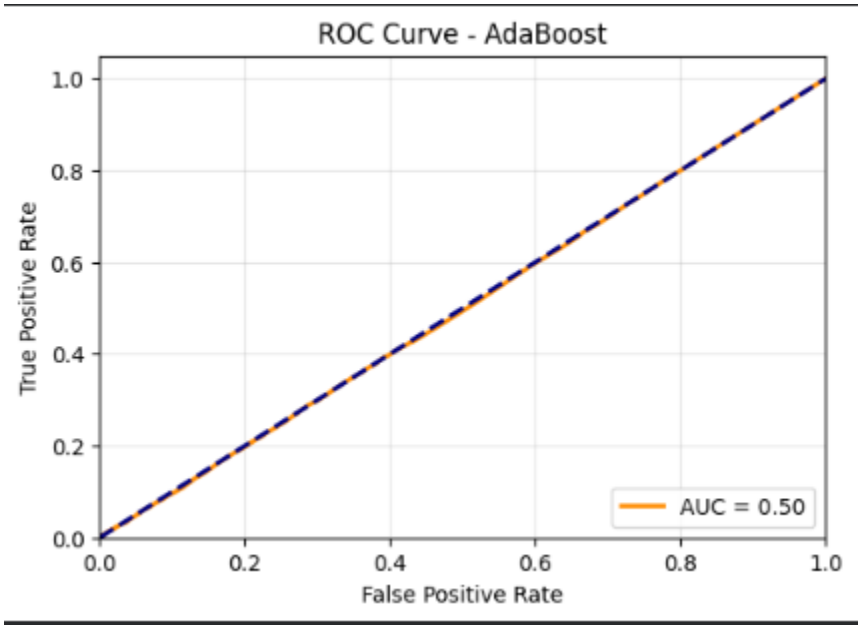
Naive Bayes



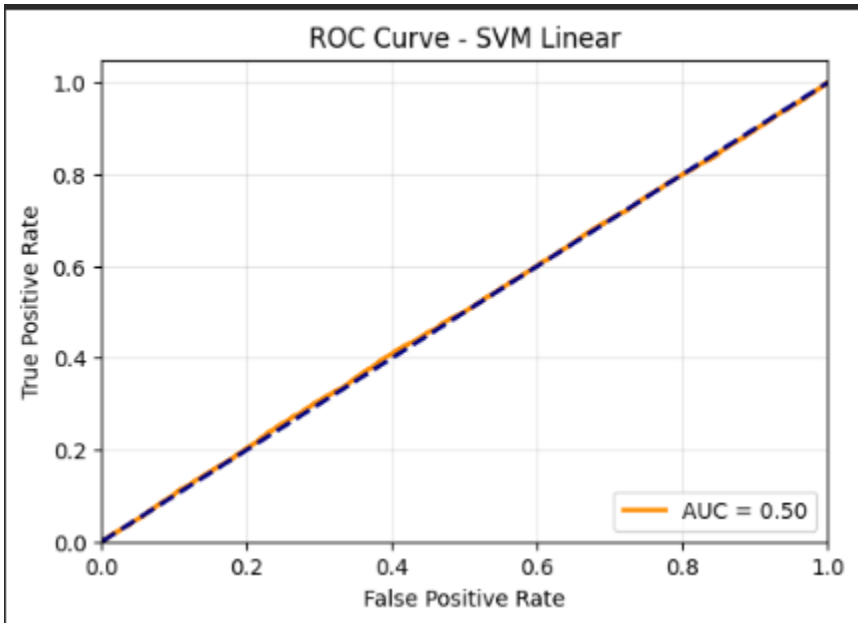
Gradient Boosting



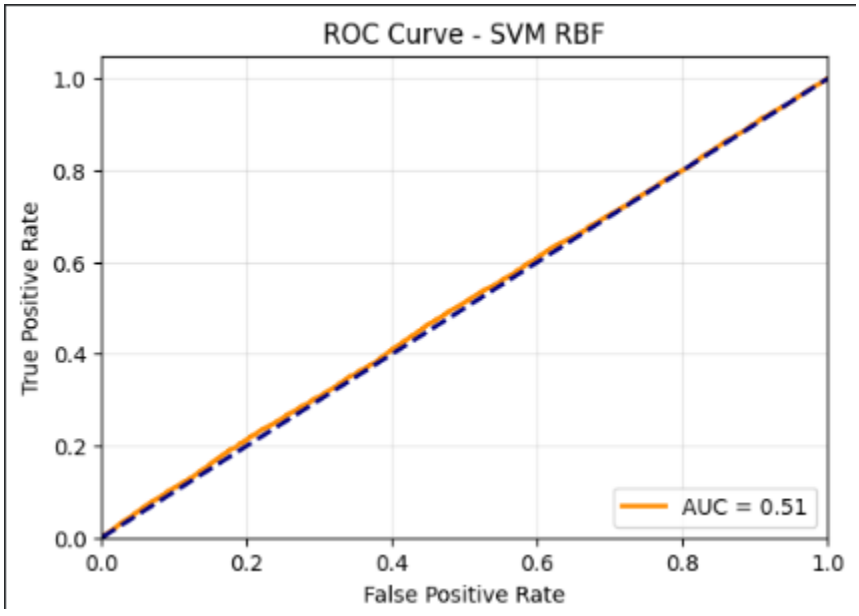
ADA Boost



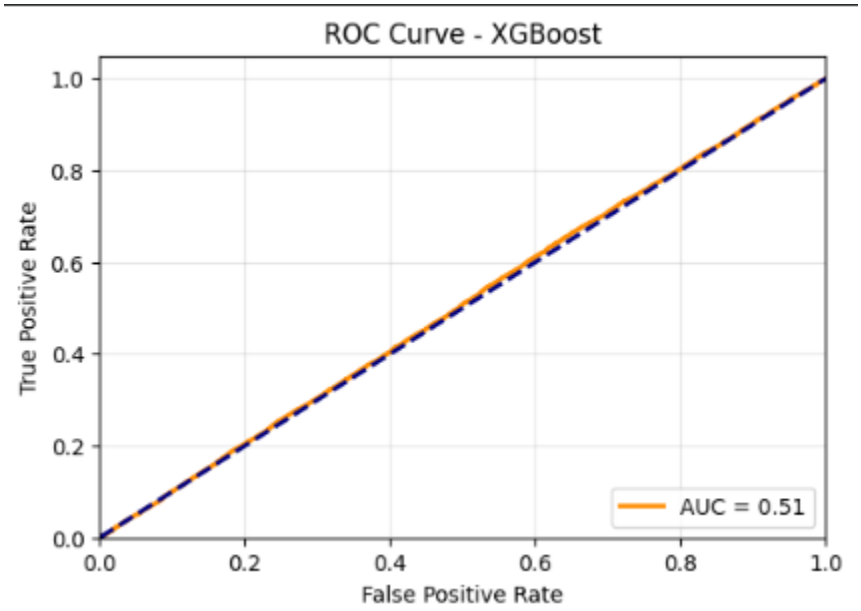
SVM linear



SVM RBF



XG Boost



Group Member Contributions:

1. Tasneem Tuhfa (24241172) (Section-04)

- Train-test split and dataset structuring
- Feature scaling (StandardScaler)
- Logistic Regression + MLP implementation
- KNN model implementation + evaluation

2. Nabilah Hossain (22299295) (Section-04)

- Data visualization (countplot, heatmap)
- Categorical encoding (label + one-hot encoding)
- Feature engineering (mental_health_index, lifestyle_balance, etc.)
- Decision Tree + Naive Bayes implementation

3. Nazifa Tahsin (22201713) (Section-03)

- Advanced preprocessing (resampling using RandomUnderSampler)
- Random Forest + Gradient Boosting models
- SVM models (Linear & RBF)
- Confusion matrix visualization + accuracy comparison

4. Nishat Tasnim Luba (24241354) (Section-01)

- Data loading and initial inspection
- Missing value analysis & basic cleaning
- XGBoost + AdaBoost models
- ROC curve, AUC, precision, recall, F1-score analysis